

1	(a)(i)	$0 = 30 \sin 30^\circ - (9.81) t$ $t = 1.529 \text{ s}$ time taken = $1.529 \times 2 = 3.06 \text{ s}$	M1
			A1
	(a)(ii)	$x = u_x t = 30 \cos 30^\circ (3.06) = 79.5 \text{ m}$	A1
	(b)(i)	A straight line between 0 and $3 \text{ s} < t < 3.1 \text{ s}$ with $v = 0$ at $t \approx 1.5 \text{ s}$ between -15 m s^{-1} and 15 m s^{-1}	M1
			A1
			A1
	(b)(ii)	curve with time taken moving upwards < time taken moving downwards (gradient at $t = 0 \text{ s}$ is approximately 9.81) approximately same area	B1
			B1

2	(a)	pressure = force / area OR weight (of liquid) / area	[B1]	
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